

Mr Corbin's 2026 Mock AMC8A

Problem 1 Gabriela is adding up the first 20 odd numbers. What is her sum?

- (A) 105 (B) 210 (C) 381 (D) 400 (E) 421

Problem 2 Mathew is volunteering afternoons at a soup kitchen every 12 days. His tenth afternoon volunteering takes place on a Thursday. What day of the week did his first afternoon volunteering take place?

- (A) Monday (B) Tuesday (C) Wednesday (D) Thursday (E) Friday

Problem 3 What is the sum of the divisors of 2026?

- (A) 1013 (B) 1015 (C) 1016 (D) 3042 (E) 5040

Problem 4 What is the sum of the mean, median, mode, and range of the data

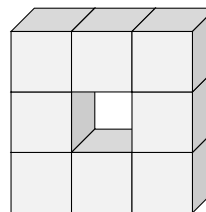
3, 1, 2, 0, 1, 4.5, 2, 3.5, 2

- (A) $173/18$ (B) $191/18$ (C) $19/9$ (D) 10 (E) $23/2$

Problem 5 Lain rolls 2 fair six-sided dice and sums up the results. Given that her sum was not a multiple of 4, what is the probability that it was even?

- (A) $1/4$ (B) $9/32$ (C) $1/3$ (D) $10/27$ (E) $1/2$

Problem 6 Steve arranges 8 unit cubes into a donut shape as shown to the right. What is the number of faces, minus the number of edges, plus the number of vertices, of his donut? (two regions joined by an edge are considered distinct faces)



- (A) -2 (B) -1 (C) 0 (D) 1 (E) 2

Problem 7 Suppose that $a \diamond b = \frac{a+b}{a \times b}$. Then what value of x solves $(2 \diamond 2) \diamond x = -2$?

- (A) $-1/3$ (B) 0 (C) $1/2$ (D) $10/7$ (E) 6

Problem 8 How many numbers from 50 to 200 inclusive are multiples of 7 or 13, but not both?

- (A) 28 (B) 29 (C) 30 (D) 31 (E) 32

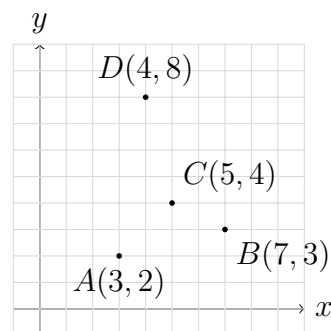
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Problem 9 Edna adds up the first 2026 even numbers, and subtracts the first 2026 odd numbers. What is the result?

- (A) -2026 (B) 0 (C) 1013 (D) 2026 (E) 4052

Problem 10 The concave quadrilateral $ABCD$ has coordinates $A = (3, 2)$, $B = (7, 3)$, $C = (5, 4)$, and $D = (4, 8)$. What is the area of $ABCD$?

- (A) 8 (B) 8.5 (C) 9 (D) 9.5 (E) 10



Problem 11 What is the units digit of $2024^{2024} + 2025^{2025} + 2026^{2026}$?

- (A) 6 (B) 7 (C) 8 (D) 9 (E) 0

Problem 12 There are 75 people at a park, 43 are walking around, and 36 are looking at their phone. What is the least number that could be doing both?

- (A) 4 (B) 6 (C) 7 (D) 35 (E) 36

Problem 13 Alice, Bob, Carol, and Daniel are sitting in a row of four chairs. They all get up, and sit in a chair different from the one they started in. How many ways can this happen?

- (A) 6 (B) 7 (C) 8 (D) 9 (E) 10

Problem 14 Comet the alien has 4 arms, 2 on each side, but no eyes. Despite their lack of eyes they still like to wear color-coordinated outfits. They have a drawer full of ambidextrous gloves that fit either a left hand or a right hand, that can be blue, green, or red. How many gloves does Comet need to take from the drawer in order to be sure they have four gloves of a single color?

- (A) 5 (B) 9 (C) 10 (D) 11 (E) 13

Problem 15 The notation $n!$ means the product $n \times (n - 1) \times (n - 2) \times \dots \times 3 \times 2 \times 1$ for any positive integer n . How many zeros are there on the right side of the digit expansion of $37!$ before you reach a nonzero digit?

- (A) 7 (B) 8 (C) 22 (D) 23 (E) 36

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Problem 16 Terence adds up all the positive integers less than 50 with an odd number of divisors. What number does he get?

- (A) 30 (B) 64 (C) 65 (D) 104 (E) 140

Problem 17 This time Terence adds up all the positive integers less than 50 that have a greatest common divisor of 1 with 50. What number does he get this time?

- (A) 500 (B) 501 (C) 1183 (D) 2025 (E) 2026

Problem 18 Qirat has one six-sided dice and one eight-sided dice. Both are fair. When she rolls them, what is the probability that the sum is less than 12?

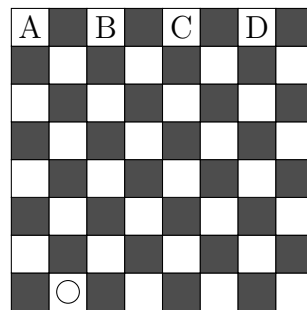
- (A) $1/8$ (B) $3/7$ (C) $1/2$ (D) $7/8$ (E) $7/48$

Problem 19 The small town of Markoville has two restaurants, Sandra's Sandwiches and Tico's Tacos. There are 200 people who eat at one of these two restaurants once a week. If someone eats at Sandra's Sandwiches then they have a 30% chance of returning to eat there again next week, and a 70% chance of going to Tico's Tacos instead next week. If they eat at Tico's Tacos, then they have a 40% chance of returning next week, and a 60% chance of going to Sandra's Sandwiches. If 100 people ate at each restaurant this week, how many do we expect to find at Sandra's Sandwiches in 2 weeks?

- (A) 16 (B) 51 (C) 93 (D) 97 (E) 117

Problem 20 On a checkerboard, a checker can only move up and to the left, or up and to the right, staying on the same color each move. Suppose that our checker starts at the bottom leftmost white square as shown, what is the probability that a checker randomly moved ends up at B after 7 moves, given that it does not end up at D ?

- (A) $7/15$ (B) $14/31$ (C) $29/62$ (D) $1/2$ (E) $33/62$



Problem 21 A sequence $a_1, a_2, \dots, a_n$ is said to be arithmetic if the difference between consecutive terms is constant, $a_k - a_{k-1} = d$ for $k = 2, 3, \dots, n$ and some number d . Given that $a_4 = 58$ and $a_7 = 49$, what is $a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + a_7$?

- (A) 58 (B) 67 (C) 398 (D) 404 (E) 406

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Problem 22 Jelani is going to buy a box of a dozen donuts for his friends board game night. He can choose any combination of 12 donuts coming in 5 different possible flavors; glazed, chocolate, sprinkles, jelly filled, or creme filled types. Jelani intends to get at least one of each type of donut. How many different configurations of boxes of donuts can Jelani bring to the board game night? (assuming the order of donuts in the box does not matter, only the amounts of each flavor)

- (A) 210 (B) 330 (C) 495 (D) 792 (E) 924

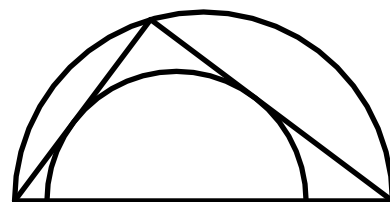
Problem 23 Phuong is painting the faces of cubes either red or blue. If two paintings of a cube are considered the same whenever one can be rotated onto the other, how many ways can she paint a cube? For example, a painting of a cube with five blue faces and one red face will be the same no matter what face gets painted red.

- (A) 6 (B) 7 (C) 8 (D) 9 (E) 10

Problem 24 Triangle ABC has area 100. Point D is on side AC so that $AD : DC = 1 : 3$. Point E is on BD so that $BE : ED = 4 : 1$. Point F is on EC so that $EF : FC = 1 : 2$. What is the area of triangle DEF ?

- (A) 3 (B) $25/6$ (C) 5 (D) 10 (E) $100/9$

Problem 25 Two half circles are drawn on the hypotenuse of a 3-4-5 triangle as shown, one being the smallest semicircle that contains the triangle with diameter on the hypotenuse, and the other being the largest semicircle contained inside the triangle with a diameter on the hypotenuse. What is the sum of the radii of the half circles?



- (A) 3 (B) $59/14$ (C) $63/14$ (D) $29/7$ (E) $9/2$

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Answer key

1. D
2. A
3. D
4. B
5. C
6. C
7. A
8. B
9. D
10. A
11. B
12. A
13. D
14. C
15. B
16. E
17. A
18. D
19. C
20. B
21. E
22. B
23. E
24. C
25. B